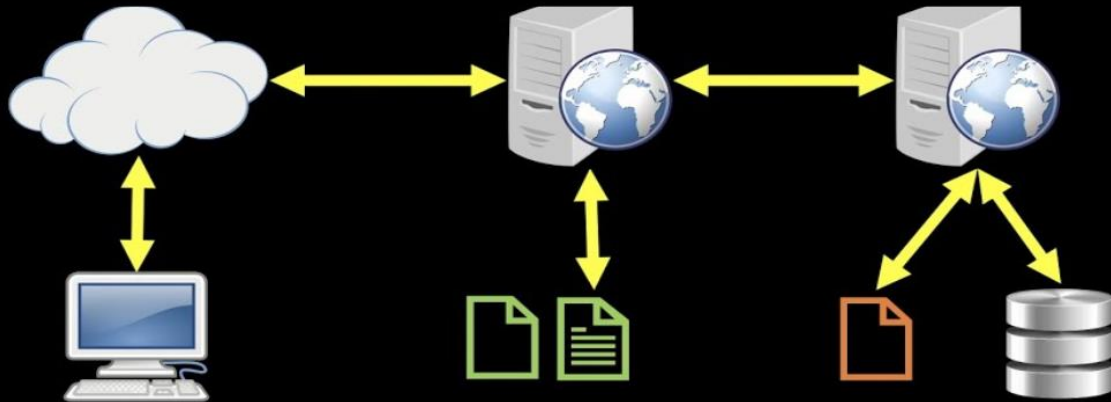


Web Server VS Application Server



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Web Server VS Application Server

What is a Web Server?

A **web server** is software or hardware that handles **HTTP requests** from clients (like browsers) and serves **static content** (like HTML, CSS, JS, images). It essentially listens to requests on a port (typically port 80 or 443 for HTTPS) and returns the requested resources.

◆ Examples of Web Servers:

- Apache HTTP Server
- Nginx
- Microsoft IIS

What is an Application Server?

An **application server** is software that provides the **business logic** for dynamic content. It processes requests, accesses databases, runs server-side code (like Java, Python, PHP), and then sends the response back to the web server or directly to the client.

◆ Examples of Application Servers:

- Apache Tomcat (Java)
- JBoss / WildFly (Java EE)
- Node.js
- Django (Python Framework running on WSGI)



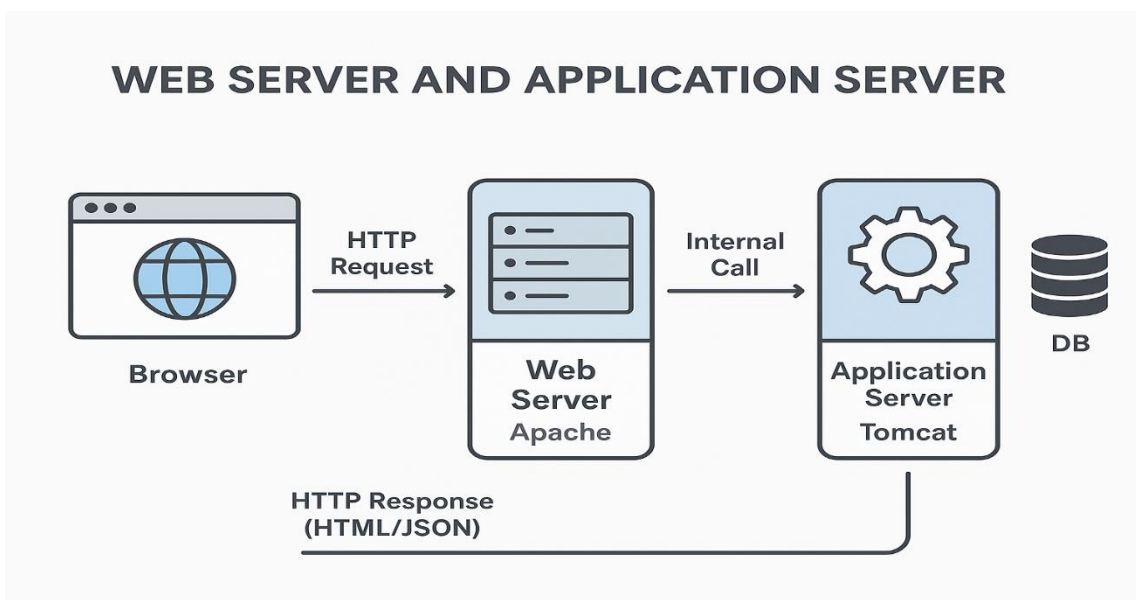


Difference Between Web Server and Application Server

Feature	Web Server	Application Server
Serves	Static content	Dynamic content (business logic)
Handles	HTTP requests only	Business logic + APIs + HTTP
Examples	Apache, Nginx	Tomcat, JBoss, Node.js
Language Support	Limited (HTML/CSS/JS)	Supports multiple languages
Can interact with DB?	No (not directly)	Yes

🔑 Example Flow:

1. **Browser** requests `www.example.com/products`
2. **Web Server (Nginx)** receives it, sees it needs dynamic content.
3. It forwards the request to **Application Server (Tomcat running a Java app)**.
4. App Server fetches data from **Database**, processes it.
5. App Server sends **HTML or JSON response** back to the Web Server.
6. Web Server sends it to the browser.



2.





Modern Stack: Nginx + Gunicorn + Django

💡 How it works:

1. **Browser** sends a request (e.g., /user/profile) via HTTPS.
2. **Nginx (Web Server):**
 - Handles the HTTPS connection.
 - Serves static files directly (CSS, JS, images).
 - For dynamic content, forwards the request to **Gunicorn**.
3. **Gunicorn (App Server):**
 - A WSGI-compliant Python HTTP server.
 - Runs your **Django** app.
 - Processes logic, interacts with DB.
 - Sends the response back to Nginx.
4. **Nginx** sends the final response to the client.

